15. Correcting for Nonstandard Conditions

Standard Versus Nonstandard Conditions

Standard conditions for ventilation design are a temperature of 70°F and an atmospheric pressure of 14.7 pounds per square inch. These standard conditions are different than those that are presently used for industrial hygiene and chemistry work (i.e., 77°F and 20°C).

In the previous design problems, charts have been used that assume that standard air is being moved through the system. These charts are not applicable to nonstandard conditions without correction. The effect of heat, pressure, and moisture in the air will change the values that are obtained from these charts.

If heat is added to the air, the molecules become more active, and the heated air expands. This relationship is stated by Charles' Law. On the other hand, as the pressure increases, the volume decreases as presented in Boyle's Law.

These two laws have been combined to form the Perfect Gas law which is stated as

$$PV = WRT \quad (2.1.7)$$

where

P = absolute pressure in pounds per square foot
V = the total volume of gas in cubic feet
W = the total mass of the gas in pounds
R = the gas constant which has a value of 53.4 for air
T = absolute temperature in °R (for temperatures in Fahrenheit (460 + F))

When humidity (water vapor) is present in the air, the partial pressures of the air and humidity together make up the total atmospheric pressure. The moisture content within the air exerts a partial pressure while the air itself exerts a partial pressure. The sum of these two partial pressures becomes the atmospheric pressure which is present for the mixture. Water vapor is then treated as a gas.

Need for Correction

Some generally accepted rules have been developed to determine when correction for temperature, pressure, and moisture content are necessary. In general, standard conditions are assumed when

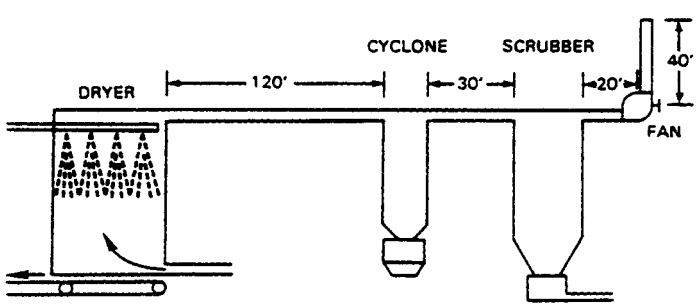
1. the temperature range is between 40°F and 100°F.
2. the pressure range is equivalent to an altitude range of -1000 feet to +1000 feet.
3. moisture content is considered standard if the air temperature is less than 100°F. If the air temperature is greater than 100°F and the moisture content is greater than 0.02 pounds of water per pound of dry air, correction is required for the moisture in the air.

Corrections are made because of the effect which these factors have on the volume of air being handled in a ventilation system. The volume of air itself remains the same regardless of the density. The weight of air moved within a ventilation system will be different and is a function of density. The pressure developed in the system will differ; and the horsepower used, as well as the rpm's at which a fan operates, will differ varying with the pressure developed within the system. The simplest way to illustrate these varying effects is to solve a problem involving nonstandard conditions.

Example Problem

A detergent manufacturer has a spray-drying operation that is used to transform liquid droplets to detergent flakes. The operation involves the upward flow of heated air at a temperature of 400°F dry bulb at a rate of 12,000 cfm. The droplets are sprayed downward through the upward flowing air, and the dried flakes collect at the bottom for further processing. An exhaust ventilation system is to be connected to the dryer to remove the moist, hot air that also contains particles of the detergent. It is determined that a high-efficiency centrifugal cleaner will be used to reclaim the particles that are exhausted, followed by a wet scrubber for final cleaning. The following presents a schematic of the required setup:

Detergent drying.



Given

Moisture removed--150 pounds/minute in dryer
At exit of wet scrubber, air is 90% saturated

Pressure drop--0.4 inches wg for dryer,
8 inches wg for centrifugal cleaner, and
15 inches wg for wet scrubber
Hood loss = 1.5 VP at entry to duct

Problem

Size duct and fan for the ventilation of the process, considering the effect of temperature and humidity.

Solution

1. To determine actual volume exhausted from spray dryer. Air exhausted contains moisture that must be considered because of the weight of the moisture and partial pressure exerted by the water vapor.

Moisture exhausted

150 lbs/minute of H_2O

Dry air exhausted at standard conditions

12,000 cfm at 400°F
12,000 cfm x 0.075 lb/ft^3 = 900 pounds/minute dry air exhausted at standard conditions

Water/pound dry air

$$\frac{\text{lbs } H_2O}{\text{lb dry air}} = \frac{150 \text{ lbs } H_2O/\text{min}}{900 \text{ lbs dry air/min}} = 0.167 \text{ lbs water/lb dry air}$$

To determine the characteristics of the air/water mixture, consult Psychrometric Chart for 400°F and 0.167 lbs H_2O.

Dew Point = 142°F
Wet bulb temperature = 153°F
Humid volume = 27 ft^3/lb dry air
Density factor = 0.57
Enthalpy = 300 BTU/lb dry air

To find actual volume of air/water mixture to be exhausted:

$Volume_{actual}$ = Humid vol x weight of dry air/minute
= 27 ft^3/lb dry air x 900 lbs dry air/minute
= 24,300 cfm

2. Calculate resistance and hood loss from dryer to cyclone.

Duct Size

Assume minimum transport velocity to be 4500 fpm

$$A = \frac{Q}{v} = \frac{24{,}000}{4{,}500} = 5.4 \text{ ft}^2$$

Using 32 inch duct (area 5.585 ft^2)

$$v = \frac{Q}{A} = \frac{24{,}300}{5.585} = 4351 \text{ fpm}$$

Correcting for density to obtain VP:

$$v = 1096.2 \sqrt{VP/\rho} \quad (2.3.1)$$

where ρ = 0.075 x density factor = 0.075 (0.57)
VP = 0.67 inches wg

Hood Loss

Since H_e is given as 1.5 VP, the hood loss (corrected for density) is obtained from:

$$\text{Loss}_H = h_e \times d$$
$$= 1.5 \times 0.67 \times 0.57$$
$$\text{Loss} = 0.57 \text{ inches wg}$$

Acceleration Factor

One VP = 0.67 inches wg

Friction Loss--Dryer to cyclone

0.65 inches wg/100 ft x 120 ft x 0.57 = 0.44 inches wg

3. Calculate resistance within cyclone.

Cyclone Loss

8 inches wg at standard x 0.57 = 4.56 inches wg

4. Calculate resistance between cyclone and wet scrubber.

0.65/100 ft x 30 ft x 0.57 = 0.11 inches wg

5. Calculate air/water mixture at exit of scrubber.

At the point the air leaves the wet scrubber, it has a different set of characteristics than were previously calculated. This is because of the humidifying effect upon the air that occurs as the air passes through the wet scrubber.

> "An important characteristic of wet collectors is their ability to humidify a gas stream. The humidification process is generally assumed to be adiabatic or without gain or loss of heat to the surroundings. Therefore, water vapor is added to the mixture, but

the enthalpy, BTU/lb dry air, remains unchanged. During the humidification process, the point on the Psychrometric Chart that describes the mixture moves to the left, along a line of constant enthalpy, toward saturation where no more water vapor can be added.

"All wet collectors do not have the same ability to humidify. If a wet collector is capable of taking an air stream to complete adiabatic saturation, it is said to have a humidifying efficiency of 100%. The efficiency of a given device to humidify may be expressed by

$$\text{Humidifying efficiency } \% = \frac{(t_i - t_o)}{(t_i - t_s)} \times 100$$

where

t_i = dry bulb temperature at inlet
t_o = dry bulb temperature at outlet
t_s = adiabatic saturation temperature

(Industrial Ventilation Manual, ACGIH, 1974, page 6-16)

To describe the characteristics of the air at the outlet of the scrubber, the above formula is used to calculate the temperature.

$$0.90 = \frac{(400 - t_o)}{(400 - 153)}$$

$$t_o = 178°F$$

Since in previous reference to Psychometric Chart
Enthalpy = 300 BTU/pound
and enthalpy remains constant, reading the Psychrometric Chart for the characteristics of the new mixture yields

Dew point 151°F
Wet bulb temperature = 153°F
Humid volume = 21.8 ft^3/pound
Enthalpy = 300 BTU/pound dry air
Density factor = 0.74

Resistance loss for the scrubber is assumed to be 15 inches wg since specifications of the manufacturer will include the effect of humidification on the pressure loss.

Exhaust volume at collector outlet

Q = humid volume x weight dry air/min
Q = 21.8 ft^3/lb x 900 lbs/minute
Q = 19,620 cfm

6. Calculate losses between collector and fan

0.44 inches wg/100 ft x 20 ft x 0.74 = 0.07 inches wg

7. Fan

Generally, if static pressure losses are less than 20 inches wg, the effect of the lower pressure in the duct as compared to atmospheric pressure is ignored. However, if pressure decreases significantly below atmospheric pressure, the air within the duct will expand. If this effect is not considered when sizing the fan, the volume of air handled will not be sufficient.

Since the static pressure losses to point of fan = 21.42 inches wg (the total of the losses from entry to the fan), it will be necessary to make this correction.

To correct for low pressure using Boyle's Law (2.1.5)

$$\frac{P_1}{P_2} = \frac{V_2}{V_1}$$

Standard atmospheric pressure = 407 inches wg

$$\frac{407 \text{ inches wg}}{(407 - 21.42) \text{ inches wg}} = \frac{Q_2}{19{,}620}$$

$$Q_2 = 20{,}710 \text{ cfm}$$

The change in pressure also affects the density of the air. To obtain the corrected density factor:

$$\frac{P_1}{P_2} = \frac{d_1}{d_2}$$

$$\frac{407 \text{ inches wg}}{385.6 \text{ inches wg}} = \frac{0.74}{d_2}$$

$$d_2 = 0.70$$

To determine the corrected VP_i:

$$v = \frac{Q}{A}$$

$$v = \frac{20{,}710}{5.585}$$

$$v = 3708 \text{ fpm}$$

Then to obtain VP_i

$$v = 1096.2\sqrt{VP_i/\rho} \quad (2.3.1)$$

where $\rho = 0.075 \times d = 0.075\ (0.74)$

$VP_i = 0.64$ inches wg

8. Duct discharge losses

The loss in the discharge section must be determined. Since the static pressure losses in the discharge section will be low, it will not be necessary to correct for pressure differences. The air volume and density will be as calculated before correction: i.e., Q = 19,620 and d = 0.74.

Loss in discharge (use 34 inch diameter duct)

0.44 inches wg/100 ft. x 40 ft x 0.74 = 0.13 inches wg

9. Select fan and rating

$$SP_f = SP_o - SP_i - VP_i \quad (2.10.7)$$
$$= 0.13 - (-21.42) - 0.61$$
$$= 20.94 \text{ inches wg}$$

Correcting for density since fan rating tables assume standard conditions:

$$\text{Fan}_{SP} = \frac{20.94}{d} = \frac{20.94}{0.74} = 28.20 \text{ inches wg}$$

Using the fan rating table below for the values:

SP = 28.30 inches wg
Q = 20,710 cfm

Table 2.15.1

CFM	22" SP		24" SP		26" SP		28" SP		30" SP		32" SP	
Volume	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
16,000	1130	74.6	1220	82.0	1305	91.5	1380	100	1450	108	1510	115
18,500	1180	81.8	1270	89.2	1355	98.7	1430	107	1500	115	1560	122
21,000	1228	88.8	1318	96.2	1403	106	1478	114	1548	122	1608	129
23,500	1274	95.6	1364	103	1449	113	1524	121	1594	129	1654	136
26,000	1318	102	1408	110	1493	119	1568	128	1638	136	1698	142
28,500	1360	109	1510	117	1535	126	1610	135	1680	142	1740	148
31,000	1400	115	1550	123	1575	133	1650	141	1720	148	1780	154
33,500	1438	121	1588	129	1613	139	1688	147	1758	154	1818	160

Interpolating between the values:

Q = 18,500 cfm and 21,000 cfm
SP = 28 inches wg and 30 inches wg

the following characteristics are obtained:

RPM = 1482
BHP = 114

Summary

When the air is other than standard, it is necessary to make corrections for the fact that the air has a different density, since the operation of a ventilation system is affected by this change. In general, as the air becomes less dense, the effect of losses as a result of turbulence and friction also becomes less. This in turn affects the size and operating characteristics of the air mover. Normally the correction is necessary for air if the temperature is greater than 100°F, if the altitude varies beyond 1000 ft from sea level, or if the moisture content of the air is greater than 0.02 pounds of water per pound of dry air.